Introduction

Revise and update the introduction. Make sure to cite your references inside the introduction. In the last paragraph, include a short summary of the following sections.

Artificial Intelligence (AI) now is quickly changing how companies/organizations work and operate by allowing machines to think, learn and make decisions like humans [1]. With the development of AI technologies like Natural Language Processing (NLP) and Machine Learning (ML), companies now have flexibility to analyze higher amounts of customer feedback faster and accurately, which this operation was impossible before with manual methods [2,3]. AI-driven sentiment analysis allows businesses to understand customer sentiment/emotions and opinions that they write in reviews and social media, simplifying faster and data-driven decision-making processes [4,5].

References:

1. Russell, S. & Norvig, P. (2020). Artificial Intelligence: A Modern Approach. [https://doi.org/10.5555/3455716]
2. Davenport, T. & Ronanki, R. (2018). The promise and challenge of AI. Harvard Business Review. [https://hbr.org/2018/01/the-promise-and-challenge-of-ai]
3. Jha, B.K., Sivasankari, G.G. & Venugopal, K.R. (2021). Sentiment Analysis for E-Commerce Products Using NLP. [https://search.proquest.com/openview/144340bf57d4edac4af745803135d5b2/1]
4. Lee, P., Chakraborty, I. & Banerjee, S. (2022). Artificial Intelligence Applications to Customer Feedback Research. [https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4126443]
5. Kasem, M.S., Hamada, M. & Taj-Eddin, I. (2024). AI in CRM: A Sentiment Analysis Approach. [https://ieeexplore.ieee.org/document/10545168]

This project mainly talks about developing an AI- powered system for Talabat, which is a popular food delivery service in Jordan, with its problem being managing and analyzing the high amounts of sentiment analysis to detect customer feedback from all platforms

such as social media/emails/application [6]. The proposed idea system will use generated actionable insights that enhances service quality and customer experience [7]. The implementation hopes that it will help in overcoming challenges like multi-language feedback processing, accuracy needs, and integration with existing systems [8,9].

References:

6. Talabat. (n.d.). Contact Us. [https://www.talabat.com/jordan/contact-us]
7. Okeke, N.I., Alabi, O.A., Igwe, A.N., Ofodile, O.C. & Ewim, C.P.M. (2024). AI in Customer Feedback Integration. [https://www.researchgate.net/publication/389680391]
8. Monterey AI. (2024). Top 9 Challenges of Traditional Sentiment Analysis. [https://www.monterey.ai/blog/top-9-challenges-of-traditional-sentiment-analysis]
9. ScienceDirect. (2024). Customers' sentiment on food delivery services: An Arabic text analysis. [https://www.sciencedirect.com/science/article/pii/S2667096824000880]

The following sections will present The Project Management Plan which includes detailing scope, objectives, milestones and resource allocation. It will also include The Time, Cost, and Resource Management Plan, including Gantt Charts and critical path analysis, also the Change Management Plan which will address user awareness, readiness, and resilience strategies. Section will provide The Risk Management Plan which will identify and mitigate potential risks, Project Recommendations and Justifications which will discuss stakeholder communication and planning decisions and the Performance Review, evaluating research methodology reliability and project planning effectiveness.

Project Management Plan

1 Scope management plan

The scope of this project is to design, develop, and implement an AI- sentiment analysis system for Talabat to automate and improve the analysis of customer feedback from all platforms. The system will collect customer reviews, messages, and social media comments, analyze their sentiment (positive, negative, neutral), and produce actionable reports to improve customer service and decision-making.

1.1 Project Requirement

1.1.1 Project Functional Requirement

1. The system will collect customer feedback from multiple platforms (app, social media, emails)
2. The system will process multi-language feedback with a focus on Arabic and English
3. The system will categorize sentiment analysis as positive, negative, or neutral.
4. The system will generate daily and weekly sentiment summary reports with visual dashboards.
5. The system will allow managers and customer support tram to filter and categorize the feedback whether is it from delivery, food quality , other.
6. The system will alert purposed teams on important negative feedback or repeated negative complaints.
7. The system will be able to export reports for further analysis

1.1.2 Project Non-Functional Requirement

1. The system is expected to reach an accuracy of 85-90% in sentiment analysis
2. The system is expected to ensure data privacy and align with relevant data protection laws.
3. The system is expected to have a response time of 5 seconds to generate summary reports
4. The system is expected to be available all the time.
5. The system is expected to handle exceeding numbers of customer feedback by keeping the same performance.
6. The system is expected to be user-friendly with dashboards and minimal training requirements.

1.2 Scope Description (What)

IN SCPOE	**OUT SCOPE**
Build and implement an AI system to analyze customer feedback	Producing new ways for customers to write their feedback for Talabat
Handle feedback in both Arabic and English languages	Changing the logistics or operation of how Talabat works, or how Talabat delivers food and manages drivers

Connect the AI system to Talabat's places where it receives feedback (app, social media, email).	The need to upgrade or buy new hardware or computers
The ability to generate reports and dashboards for managers	Adding other Ai tools to help customers like chatbot or product suggestions
Send alerts when the negative feedback is serious or a certain problem is repeated.	Marketing or promotional management
Train staff to use the new AI tool	Fully relying on AI without human oversight

1.3 Project Aim

The purpose of this project is to design, develop and implement AI-powered sentiment analysis systems especially for Talabat to manage and analyze customer feedback from all platforms, including apps, social media, and email. With this AI system, it will allow the company automatically to categorize feedback as positive, negative, or neutral , providing insights into customer satisfaction and repeating issues. The project aims to improve Talabat's ability to respond quickly to customer problems by the reports and alerts generated by the AI system that will support better decision making. By also supporting multi-language processing , focusing on Arabic and English, the system will understand the diverse customer base. The overall goal is to improve customer experience, increase operational efficiency in handling feedback and maintain Talabat's competitive edge within the food market by leveling up with advanced technology for its customers.

1.4 Objectives (Why)

Your specific objectives should be covering all project aspects. For example: project productivity, stakeholder satisfaction and expectation, time and financial aspects.

PROJECT OBJECTIVE	**PROJECT BENEFIT**	**SUCCESS METRIC**

Develop a reliable AI system for sentiment analysis	Feedback analysis is done automatically which will reduce manual effort	System achieves more than 85% accuracy in sentiment detection.
Support multi-language feedback processing	Understanding customers better in both languages	Similar good analysis results for both languages
Implementing the AI system with Talabat's current tools	No need to purchase any new hardware , make it easy to start using without problems	No major technical problems during setup
Creating user friendly dashboards and reports	Better decision making for managers + faster	Reports are generated daily or weekly as planned
Keeping customer data safe and following privacy laws	The trust of the customer, protecting customer and avoiding legal problems	No data breaches and pass privacy checks
Finishing the project on time and within the budget	Benefiting from the AI system on time and without overspending	Complete within 4-6 months and budget within 15,000-50,000

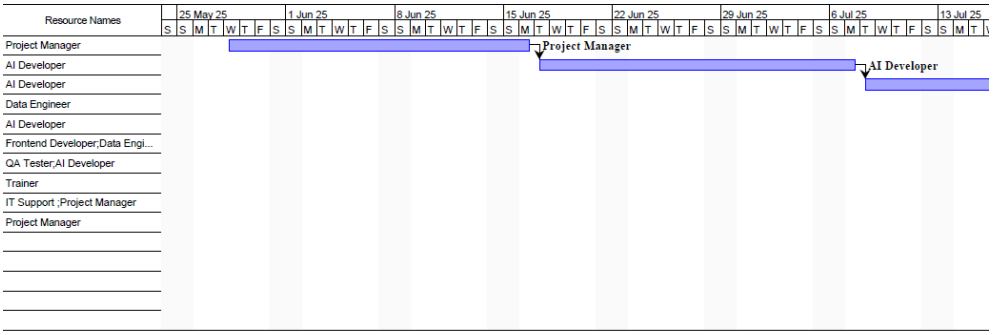
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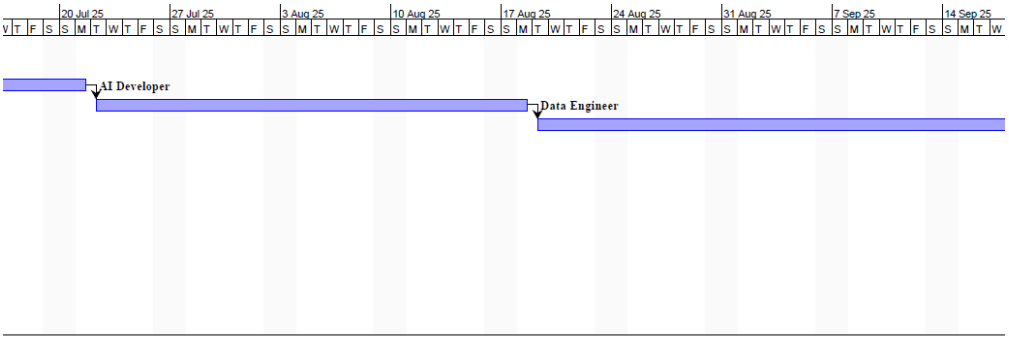
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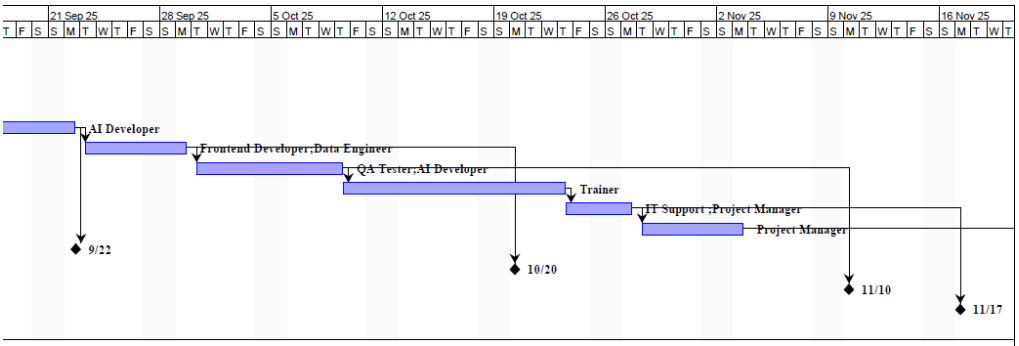
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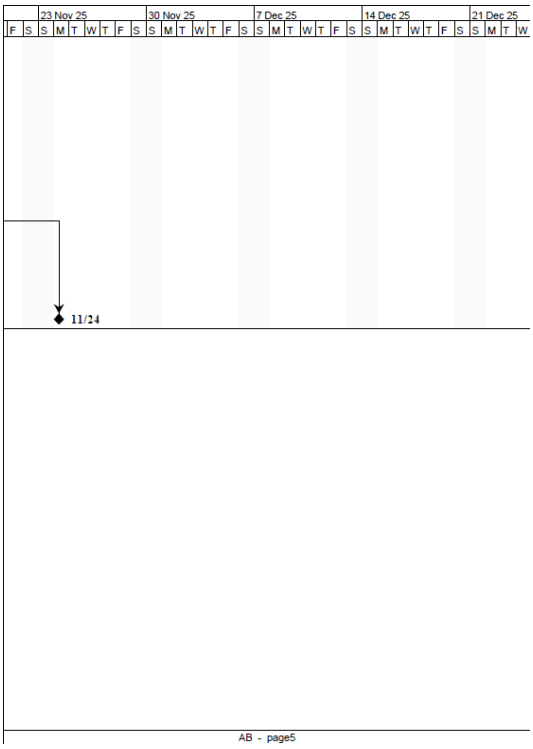
Note that prices are in JOD

		Name	Duration	Start	Finish	Predecessors	Cost
1		Project planning and gatheri...	14 days	5/28/25, 8:00AM	6/16/25, 5:00PM		$1400.00
2		Research AI tools and techni...	15 days	6/17/25, 8:00AM	7/7/25, 5:00PM	1	$1170.00
3		Design the system and its str...	10 days	7/8/25, 8:00AM	7/21/25, 5:00PM	2	$780.00
4		Develop the data collection p...	20 days	7/22/25, 8:00AM	8/18/25, 5:00PM	3	$1280.00
5		Build the sentiment analysis ...	25 days	8/19/25, 8:00AM	9/22/25, 5:00PM	4	$1950.00
6		Create dashboards and reports	5 days	9/23/25, 8:00AM	9/29/25, 5:00PM	5	$640.00
7		Integrate and test the system	7.5 days	9/30/25, 8:00AM	10/9/25, 1:00PM	6	$960.00
8		Train staff and prepare docu...	10 days	10/9/25, 1:00PM	10/23/25, 1:00PM	7	$430.00
9		Deploy the system and go live	2.5 days	10/23/25, 1:00PM	10/27/25, 5:00PM	8	$357.50
10		Close the project and review...	5 days	10/28/25, 8:00AM	11/3/25, 5:00PM	9	$500.00
11		Milestone 1 - Model Ready	0 days	9/22/25, 5:00PM	9/22/25, 5:00PM	5	$0.00
12		Milestone 2 - Dashboards Rea...	0 days	10/20/25, 8:00AM	10/20/25, 8:00AM	6	$0.00
13		Milestone 3 - Testing Complete	0 days	11/10/25, 8:00AM	11/10/25, 8:00AM	7	$0.00
14		Milestone 4 - Deployment	0 days	11/17/25, 8:00AM	11/17/25, 8:00AM	9	$0.00
15		Milestone 5 - Project Complete	0 days	11/24/25, 8:00AM	11/24/25, 8:00AM	10	$0.00

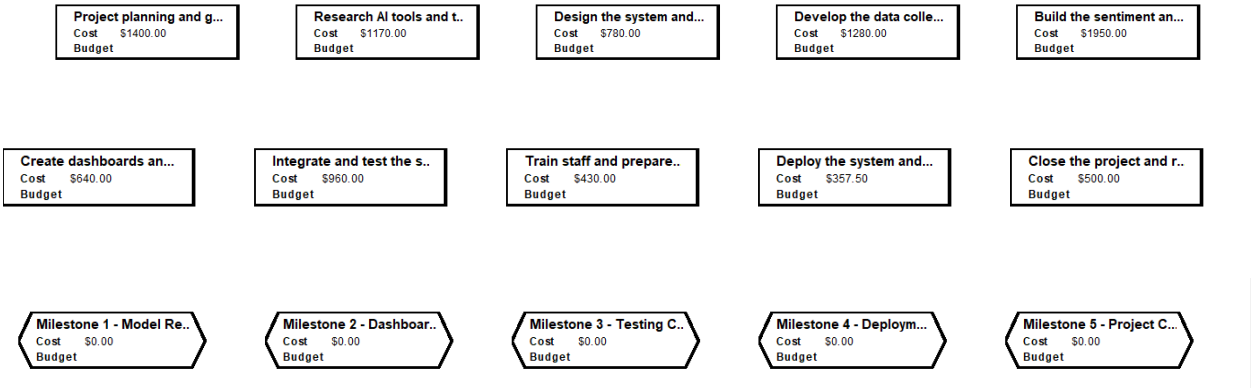



20 Jul 25
27 Jul 25
3 Aug 25
10 Aug 25
17 Aug 25
24 Aug 25
31 Aug 25
7 Sep 25
14 Sep 25
AI Developer
Data Engineer

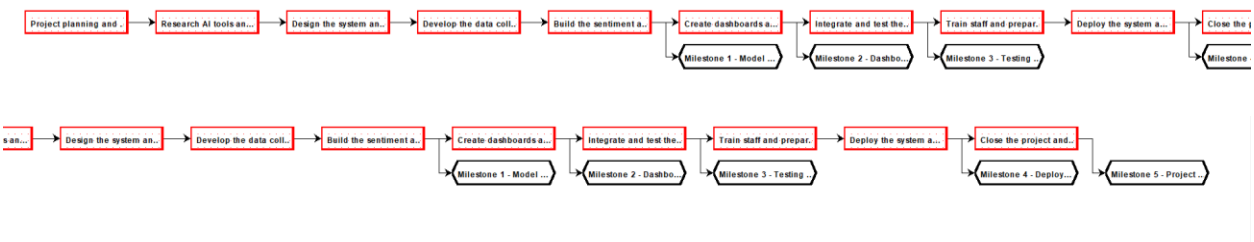

21 Sep 25
28 Sep 25
5 Oct 25
12 Oct 25
19 Oct 25
26 Oct 25
2 Nov 25
9 Nov 25
16 Nov 25
AI Developer
Frontend Developer;Data Engineer
QA Tester;AI Developer
Trainer
IT Support ;Project Manager
Project Manager
9/22
10/20
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11/17


23 Nov 25
30 Nov 25
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Build the sentiment a...
Create dashboards a...
Integrate and test the...
Train staff and prepar.
Deploy the system a...
Milestone 1 - Model ...
Milestone 2 - Dashbo...
Milestone 3 - Testing ...
Design the system an...
Develop the data coll...
Build the sentiment a...
Create dashboards a...
Integrate and test the...
Train staff and prepar.
Deploy the system a...
Close the project and...
Milestone 1 - Model ...
Milestone 2 - Dashbo...
Milestone 3 - Testing ...
Milestone 4 - Deploy...
Milestone 5 - Project ...

1.6 Milestones (When) at least three milestones

DATE	MILESTONE	DESCRIPTION
22/09	Milestone 1 – Model Ready	The AI sentiment analysis system is fully developed and ready for initial testing. The model can start analyzing customer feedback data to check sentiment accurately.
20/10	Milestone 2 – Dashboards Ready	The development of dashboards and reporting tools is completed. The dashboards will make sure that managers and staff have access to easily view and understand customer feedback summaries and insights.
10/11	Milestone 3 – Testing Complete	All parts of the AI system have been tested together to make sure everything works properly. Bugs and issues have been fixed if there's any, and the system is stable and reliable for usage.
17/11	Milestone 4 – Deployment	The AI sentiment analysis system is deployed at Talabat and l in the working environment. Staff can use it to monitor and improve customer service.
24/11	Milestone 5 – Project Complete	The project is officially done after a final review. All objectives have been fulfilled, documentation is finished. The system is fully operational and handed over to Talabt.

2 Time, Cost, and Resources Management Plan

2.1 Gannt Chart

You need to discuss all parts of the Gant Chart and the relationship between them (time, cost, resources). Also, you are required to add the milestones of section 1.1.5 into the Gantt Chart.

Part 1: General Parts of a Gantt Chart

A Gantt chart is an important project management tool that shows a project's schedule. The Gantt chart breaks down into separate tasks, every task represents part of the work that needs to be done. For every task, the chart has a duration, which is the amount of time each task will take, along with the start and finish dates that show when every task

will begin and finish. Key features of a Gantt chart that it has dependencies, which are the relationships between tasks that makes sure tasks start at the right time after specified tasks. This ordering makes sure no conflicts happen and keeps the project on track. The chart also shows resources, which refers to people assigned to each task. With resources being shown, managers can balance the work with members and avoid overloading them. Another important chart is cost, which is taken from the resources rates and the amount of time they spend working on tasks. With the costs showing along in the chart, it will allow the managers to monitor the project budget effectively. Lastly, milestones are included in the chart to mark important events or deliverables that happen throughout the project.

Part 2: Our Project's Tasks, Purpose, Dependencies, Resources, and Costs

Task Name	Purpose	Dependencies	Resource Assigned	Estimated Cost
Project Planning	Defining project objectives/goals, gather requirements, plan for the timeline and budget	None	Project Manager	2,400
Research AI tools & techniques	Exploring AI tools to see which one is the best that suits the system	Planning	AI Developer	5,070
Design system	Creating the overall AI system and data flow design	Research	AI developer	3,100

Develop data collection part	Build modules that collect feedback from all platforms	Design	Data Engineer	1,280
Build sentiment analysis model	Develop and train AI models to analyze customer sentiment from feedback	Data collection	AI developer	7,750
Create dashboards & reports	Managers have easy access to view feedback summaries by building dashboards.	Data collection	Frontend Developer	1,280
Test System	Combine all parts of the system al perform system testing to ensure every functionality is ready to be used	Model, Dashboard	QA Tester, AI developer	3,975
Train Staff	Prepare the ways of training for employees and do training sessions for Talabat's staff	Testing	Trainer	430

Deploy System	Launch the AI system in Talabat's working environment	Testing	IT Support, Project Manager	215
Close Project	Reviewing the final overview of the project, and the formal closing of the project	Deployment	Project Manager	2,400

Part 3: Our Project Milestones:

For clear tracking during the project, 5 key milestones were integrated inside the project timeline. The first milestone is Model Ready on the 22nd of Sept, it's when the AI sentiment analysis model is done after collecting data and requirements and finding the right model for the existing system. The second milestone, Dashboards Ready on the 20th of Oct, when dashboards and reporting tools are finished, which are essential for easy understanding of customer feedback and for managers to access. The third milestone, Testing Complete on the 10th of Nov, shows that the system with the integration of components is completed, confirming system stability. The fourth milestone, Deployment on the 17th of Nov, which shows that the AI system is ready to be used in Talabat's environment. The final milestone, Project completion on the 24th of Nov, marks the official ending of the project after reviewing it.

Part 4: Relationship between Time, Cost, Resources in Our Project

In the Gantt chart for the AI sentiment analysis project, time, cost, and resources are connected to each other and affect each other in different ways:

- Time and Resources : The time each task takes depends heavily on the resource it has been assigned for. For example, having an expert AI developer working full time helps completing complex tasks like finishing the AI models in the time it needs. If fewer resources are available, or not skilled enough, tasks may take longer which leads to delaying the project timeline not the same as the managers expected it to be.
- Resource and Cost: Each resource has a daily cost based on their salary rates in Jordan. If higher prices are paid for resources to tasks, it will lead to an increase in the overall project cost. The right way is to effectively use the resources at the right price, making sure the right people work on the right tasks which will help control costs.
- Time and Cost: Opposite relationship occurs between time and cost. When the project timeline is shortened, it means more resources are needed which leads to an increase in costs, while reducing costs by limiting resources will lead to extending the timeline of the project because of the added time needed to finish the tasks. The project plan balanced these to meet the deadline of Nov 24th, 2025, within the specified budget of 50,000.
- Milestones : Metting milestones as critical achievements in time and budget means that resources are used efficiently and costs are on track. Delays at milestones means time needs to be adjusted, cost, or resources , to put the project back on track.

2.2 Critical Path Analysis

You are required to specify the critical path of your project by listing what are the tasks that fall on the critical path and what is the relationship between the critical path and the project deadline. Finally, you should mention what your procedures are to guarantee there will be no delay on any of the critical path tasks.

The critical path is the longest path of dependent tasks that determines the shortest time to finish the project. Any delay that happens in the tasks in the path will directly cause a

delay in the overall project completion. Based on our project schedule , the following tasks are on the critical path:

1. Project planning and gathering requirements
2. Research AI tools and techniques
3. Design the system and its structure
4. Develop the data collection part
5. Build the sentiment analysis model
6. Create dashboards and reports
7. Integrate and test the system
8. Train staff and prepare documentation
9. Deploying the system and going live
10. Close the project and review results

Each task is dependent on each other, which means that each task must be completed before the one after it begins, creating the longest duration path through the project.

Relationship Between Critical Path and Project Deadline:

The project deadline is 24th Nov 2025, which is the end of the last task on the critical path. The critical path length directly sets the minimum time needed to complete the project. If any task in the path is delayed, the whole project deadline will change, making it essential to manage these tasks carefully.

Procedures to Prevent Delays on Critical Path Tasks:

To make sure no delays happen on the tasks in the critical path, the project will do the following procedures:

- Careful Monitoring: Consistent updates with progress tracking on the task status, to identify if there's any delays early.
- Resource Management: Making sure the tasks are assigned to the specified resource, and enough people are assigned for each task.
- Risk Management: Plan for risks that may happen and brainstorm ways to avoid or handle them throughout the project
- Contingency Planning: Add extra time and resources for the project ready in case any unexpected problem occurs in the project, so the finish can still be finished on time.
- Effective Communication: Keep everyone inside the project informed with every crucial update and talking to each other so that the problem gets solved quickly without any delays.
- Progress Review: Assign meetings to check progress for each task, discuss and solve problems, and adjust plans if necessary to stay on time.

2.3 Produce A Cost Estimate (Budget)

You should use the same template as (topic 12, slide 41). Use the Excel template.

Note: Prices are in JOD

	A	B	C	D	E	F	G	H	I	J
1					Count	Unit Hours	Cost/Hour	Subtotals	Level 1 Totals	% of Total
2	**Project Management**								$9,280	20%
3	Project Manager				1	172	$13	$2,150		
4	AI Developer				1	460	$10	$4,600		
5	Data Engineer				1	200	$8	$1,600		
6	Frontend Developer				1	40	$8	$320		
7	QA Tester				1	60	$6	360		
8	Trainers				0.5	80	$5	$200		
9	It Support				0.5	20	5	$50		
10										
11	**Hardware**								$2,000	4%
12	Computers				5		$200	$1,000		
13	Servers				1		$1,000	$1,000		
14										
15	**3.0 Software**								$10,000	22.08480565
16	3.1 AI tools & Cloud APIs							$6,000		
17	3.2 Specialized Licenses (if any)							$2,000		
18	3.3 Outsourced Testing							$2,000		
19										
20										
21	**4.0 Environmental Project**								$5,000	11.04240283
22	4.2 Deployment Support							$2,500		
23	4.3 Staff Training & Documentation							$2,500		
24										
25	**Miscellaneous & Contingency**								$19,000	41.96113074
26	Contingency Reserve							$4,000		
27	QA Testing & Unexpected Costs							$5,000		
28	Project Management Buffer							$10,000		
29		**Total Project Cost Estimate**							$45,280	

3 Change Management Plan

3.1 Awareness

List your procedures to increase the awareness of users of the coming change and provide updates on the scope, timing, and benefits of the program.

To fully introduce the new AI sentiment analysis at Talabat, it's important to build awareness among all staff about the new change. Awareness includes communicating about the scope, purpose, timeline, and benefits of the system to reduce uncertainty and resistance.

- Meeting and Presentations: Do sessions and meetings where the leaders of the project clarify the system's goals, importance, how it will change daily work, and the expected benefits, which is the faster processing of customer feedback, improved decision making, and better customer satisfaction. The purpose of the meetings is to allow staff to ask questions and express their concerns regarding the system.

- Consistent Email Updates: Sending weekly emails expressing progress, the project has been done, upcoming milestones, and any unexpected changes in scope or schedule which makes the staff engaged and knows what will happen next to be prepared.
- Guides and Most Asked Questions: Prepare documents that explain the new AI system model features and advantages. Documents will be available through the company internet or email to discuss any common asked questions.
- Announcements and Posts: Using internal communication to share updated information, sharing success stories using the AI model, and motivational messages to keep positive momentum.
- Manager Communication: A manager that communicates effectively with their teams, providing training, answering immediate questions.

3.2 Readiness

List your procedures that help the users feel confident and discuss how they will be able to use the new tool / process / system.

Making sure that staff are feeling prepared and confident to use the new AI tool is important for smooth change and high adoption rates

- Training Sessions: Organizing specific workshops where users learn how to use the AI tool, for example, customer service agents get training on how to extract sentiment reports from the system, while managers learn how to extract dashboards to help them with decision making.
- User Manuals and Video Tutorials: Creating step-by-step guides for users along with videos that users can watch them any time for extra help using the tool. These resources help strengthen the user usage of the tool and support different learning styles.
- Practice Environment: Provide a test version of the system that the users can test it and practice without effecting the real data showing. By doing this, users will practice using it and reduce anxiety and allow the user to experience the system and recover from mistakes.

3.3 Resilience

List your procedure to prepare for launch and support the users after go-live to help with the transition.

Preparing for launching and staying with the users after going live ensures the new system is capable of being beneficial and continues to give value

- Dedicated Help: prepare support team available on phone, email or chat, that assists users if they face any problems, questions, feedback. With quick response, it builds trust and reduces anxiety.
- Continuous Feedback : Collect staff feedback through surveys, meetings, or system analysis. With this information, it will be used to identify problem points and gives priority for fixes and improvements
- Training: additional training sessions in extra time to discuss common issues or questions. This helps users collect more information about the system and understand it better and deepen their confidence.
- System Updates: Inform users about updates, new capabilities, or performance improvements to let them stay in knowing every part of the system.
- Open Communication: Any user that feels to open up and criticizes part of the system, supportive environment to discuss issues and suggest ideas. This collaboration will help enhance the system.

4 Risk Management Plan

4.1 Risk Types (at least three risks)

Risk Type	Risk name	Scenario	Mitigation Plan
R1	Incorrect Sentiment Detection,	The AI model may misunderstand customer feedback (multi versions of Arabic , sarcasm)	Prepare high quality training for the AI model, especially for Arabic language , let human oversight for double check

R2	Trouble Connecting Systems	The new AI system may not be compatible with Talabat's current infrastructure	Test the AI system early and work closely with Talabat's IT team.
R3	Breaking Privacy Rules	If customer data is not handled the way it should be, data leaks or breaches may happen which will break privacy laws	Keep data anonymous and follow all data regulation laws needed, with regular checks on the system.
R4	Staff don't accept the system	Employees may not trust or cannot be adapted to the new AI system	Prepare intense training showing the importance of the AI system and how it makes their job easier
R5	Spending money on the budget	The project could cost more than what managers planned and offered.	Prepare extra money aside in case and keep track of all costs regularly.

4.2 Probability chart:

Probability	**Description**		
	Risk name	**Qualitative**	**Quantitative (if measurable)**
Low	R1 – Wrong Sentiment Results	Wrong sentiment analysis may happen with tricky feedback/language (sarcasm)	Less than 20% chance.
	R4 – Staff	Most staff likely will likely accept the system after	Less than 20% chance

	don't accept the system	knowing its benefits	
Medium	R2 - Trouble Connection on Systems	Could happen when connecting the systems if not handled properly	Around 40-50% chance
	R5 – Spending too much money	Costs might go above the budget without control	Around 40-60% chance
High	R3 – Breaking Privacy Rules	Very likely unless its carefully managed, strong rules and checks needed	More than 60% chance

4.3 Impact chart:

Impact

	Risk name	Impact Description
Low	R3 – Breaking Privacy Rules	It will lead to the loss of customer trust but won't stop the system from doing its core functionality.
	R4 – Staff Don't Accept the System	It will slow down the process of adoption and using the system effectively, but the system will keep running
Medium	R2 – Trouble Connecting Systems	The problem could delay the launch of the system if the system doesn't work well with Talabat's platforms
High	R1 – Wrong Sentiment Results	The main job of the system is working incorrectly, it directly affects service quality, reports ,and decision making
	R5- Spending	When spending money over the budget, project may lead to features

	Too Much Money	being cut out, or the project being delayed.

4.4 Probability and Impact Matrix:

Probability/Impact Matrix				
		Impact		
		Low	**Medium**	**High**
Probability	**High**	R3 – Privacy Risk		
	Medium		R2 – Integration Risk	R5 – Budget Overrun
	Low	R4 – Staff Not Accepting The System		R1 – Wrong Sentiment Results

5 Software development methodology

You need to mention your selection for the project management methodology (e.g., Waterfall, Agile, Rapid Application Development (RAD) and explain how you will utilize it in your project.

For this project, Agile methodology is selected as the most suitable method. In Agile, the focus is on iterative development, frequent feedback, and flexibility, which aligns very well with the needing of developing an AI sentiment analysis system. The selection of Agile came

because normally AI projects need changes as new data insights or technical problems appear. Agile allows adapting requirements during the development cycle [1]. Agile assures on delivering small, working parts of the system, which is called sprints, which allows early testing and validation. Agile supports the idea of consistently communicating with Talabat's managers and IT staff, ensuring the system meets their real needs [2]. With Agile, regular reviews with each sprint help identify early errors, reducing the risk of large failures at the end of the project.

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1. Highsmith, J. (2009). Agile Project Management: Creating Innovative Products. 2nd ed. Addison-Wesley.
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How Agile will be used in this project ?

The project will be broken down into 2–3-week phases, each phase focusing on developing a key feature of the system, such as data collection, sentiment analysis model, or reports/dashboards interface. At the end of each phase, the team will collect the working component from Talabat's stakeholders for feedback to make sure everything aligns with their requirements/needs. After gathering feedback, the team will know what priorities and features to apply for next phase, allowing flexibility to improve accuracy. Daily meetings will be held to discuss progress, problems, and tasks assigned among the development team. Continuous integration and testing will take place to ensure that new features don't corrupt existing functionality and maintain system reliability. With the system, user training and documentation will be done to prepare Talabat's staff for an easy change. The iterative method allows detection of errors and adapts to changing business needs, which reduces risk and gives the chance of the project succeeding.

Project Recommendations and Justifications

1. Methods and mediums to communicate with the stakeholders

Explore the techniques and channels employed for effective communication with diverse stakeholders. Your discussion should include a minimum of two internal and two external stakeholders, encompassing both technical and non-technical individuals. Examples of

methods and mediums include written documents, reports, online meetings, in-person meetings, or presentations. Explain how project research and the intended audience will influence the method and medium. Note: To respond to this section, please complete the provided table.

Stakeholder name/title	Stakeholder role	Stakeholder type	Methods or mediums	Frequency	Justification
Talabat Manager	Decision Maker, responsible for the project success	Internal	Face to face meetings, progress reports	Weekly	The manager needs to see regular updates on the project to make sure goals are being fulfilled. Managers need to assure budget, and system quality, and their requirements are met. Face-to-face meetings will allow further discussions.

IT or Customer Support Employee	System user, implementation support	Internal	Training session inside the company , online, email updates	Weekly or as needed	Employees need step by step procedures to do about the system, communication also about the system updates, and troubleshooting to make sure adoption process is being held well.
Customer	Provides the feedback (data) , end user	External	Public surveys, social media polls, in-app messages, email updates	After launch	Customers affect the system's success indirectly by providing their feedback, informing them builds trust and making them know that their feedback is essential.

Government Agencies	Makes sure that the company follows legal compliance	External	Formal reports, email communication	The start of the project, at the end	The system needs to be following privacy laws, has clear documentation which makes it avoid legal problems and maintains Talabat's compliance record.

2. Arguments of Planning decisions

Present the arguments for the planning decisions made when developing the project plans. Arguments should include justifications for key points from cost estimate analysis, deliverables, success metrics, and impact analysis. Justify the project management methodology (e.g., Waterfall, Agile, Rapid Application Development (RAD)) that you selected for your project.

Many important decisions were made for planning the project during the development for the implementation of the AI-powered sentiment analysis system at Talabat. Decisions were made based on detailed analysis of cost estimates, success metrics, impact assessment, and the method used for the project management methodology.

- Cost Estimate Analysis: The project was fully estimated at 45,280 JOD, which is within the range managers expected that reached 50,000 JOD. Personal costs which are salaries for every expert around 9,280 JOD covering all important roles like AI developers, project managers, data engineers, testers, trainers, and IT support. Other costs include important items like licenses, cloud computing, data preparation, training, hardware, and contingency funds. This detailed budget shows that the project is capable if being done financially and can cover all aspects of expenses without overspending. A contingency of 19,000 JOD was included; the main purpose is to handle any unexpected challenges that may happen.
- Deliverables and Success Metrics: The project is organized around specific deliverables with structured milestones like: Model ready for testing, Dashboards and reports completed, System integration with Talabat system and testing, Staff training completed, Full deployment of the system and project closure. These milestones align with Talabat's expectations of the project to enhance customer feedback which will improve decision-making. Success will be decided based on the accuracy of the system, the target is 85-90% accuracy of sentiment analysis, the timeline of completion for the project between 4-6 months, staff adoption and satisfaction for the new change based on training and feedback, and the ability for the system to handle Arabic and language with its versions and generate reports.
- Impact Analysis: The project came with many challenges that are addressed and solved like the manual handling of the high amounts of customer feedback on many platforms, the repeated unsolved service issues, maintaining a competitive edge through enhancing customer satisfaction. The new system will lower manual effort and speed the feedback process, enable faster decision making, has plans to mitigate risks regarding technical and organizational, and change management risks to lower the negative impact on the organization. The management plan includes a strategy that ensures staff readiness and acceptance, reducing the resistance of the new system.
- Justification of Project Management Methodology: The agile method was chosen for the methodology that fits the best for the AI project due to its flexibility and iterative run. AI development projects include many challenges that could occur as

data quality issues, model tuning, and integration challenges, which agile allows the adapting to these changes quickly. With agile, the project can be broken down into parts or phases, delivering each working part of the system early, which we can gather feedback from Talabat stakeholders. By making the project by phases, it reduces the chances of late discovery of risks and major issues and improves the overall product quality. Regular meetings and continuous integration allows the project to address all problems that could be faced. Agile also supports better stakeholder communication and change management, which is essential for the organizational impact of the AI system.

3. Project recommendations

Assess the extent to which the project recommendations meet the needs of the identified organization, including fully supported rationales for planning decisions made. Your assessment may include how the recommended budget, timeline, risks, resources, and changing management plan meet the needs of the organization.

The project recommendations fully meet the needs of Talabat as a popular food delivery company in Jordan that receives much customer feedback daily. The main objective of the project is to help Talabat understand the needs of the customer then act on their feedback and more accurately using an AI-powered sentiment analysis system. Here's how every part of the plan meets the needs of the company:

- Budget: The budget recommended for this project reaches JOD 50,000, which is a logical number that matches what Talabat's managers mentioned that they are capable of investing, the cost includes software development, system testing, integration with existing tools, staff training. This means the system can be finished without the need for more money or hardware, making it realistic and affordable for a big company like Talabat.
- Timeline: The project is planned to be finished within a time between 4-6 months, with clear milestones including completing the AI model, building dashboards, testing the system, and full deployment. This organized timeline helps Talabat improve faster and start benefiting from the system within the same year, which makes it stay ahead of all competitors.

- Risk Management: Many risks within the process were carefully analyzed, which include mistakes in feedback understanding, integration problems that could happen with Talabat's system, data privacy which could lead to breaches of customer data, staff resistance or not accepting the system leading to delay in benefiting from the system, budget concerns by spending more than needed. Each risk comes with a clear plan to mitigate it and manage it, such as better training for staff, and setting aside money for backup. This shows that the plan is realistic and prepared for any challenges that could happen on the way.
- Resources: The project team working on the AI system includes AI developer, Data engineer, front-end developer, tester, trainer. Tasks leading to finishing the project are assigned to the right expert, with cost and time planned for each task based on local salaries in Jordan. No extra hardware is needed for the system because Talabat already uses advanced IT systems. With these resources, it means the project can be done with the available people and tools.
- Change Management Plan: A fully prepared change management strategy is ready, that includes awareness with meetings, emails, and internal posts to clarify the new change to the staff, Readiness with staff training, practice environments and manuals, Resilience with regular support and feedback collection with staff to keep track. This helps employees accept and use the system smoothly, which is essential for the project's success.

  Overall, the project recommendations fulfill Talabat's real needs, where it solves their main problem by not handling too many feedback manually, respecting their preferred budget and tools, project is planned to finish on time and effectively, it trains staff and prepares them for the new change which will make their job easier, including ways to reduce risks and fix concerns fast. With this plan, Talabat can improve its customer service, make smarter decisions, and stay competitive with other companies in the food delivery market.

Performance Review

1. Accuracy and reliability of the Research Methodology

Discuss the accuracy and reliability of the different research methods applied by differentiating
between the different methodologies, how the selected methodology is the best for the selected
organization, and the project circumstances. Explain the relation between the selected methodologies, sample size, and sample characteristics. Reflect on the findings of your research and how those findings can impact on the quality of your decisions and the accuracy of your conclusions (Re-evaluate the generated knowledge described in Report 1).

In this project, two main research methods were used to get and analyze data about the idea of implementing AI sentiment analysis, and customer feedback systems which are, primary research : which a survey that have been shared with Talabat's managers and employees, Secondary Research: by reviewing published articles, reports, and any information related to AI sentiment analysis and its implementation, and customer feedback systems.

- Different Methodologies and Their Accuracy & Reliability
    - Primary Research (Survey) : The survey included both types of questions, open ended and close ended for quantitative data, and open ended to get detailed qualitative insights.
        - Sampling: The sampling method used is the purposive method, meaning only the people involved directly in the work related to customer feedback or decision making in Talabat are the ones that were asked. This makes sure that only people who understand the problem were targeted and asked questions.
        - Reliability: To make sure reliability applies in the survey, the survey included two questions that asked about similar topics in different forms, helping to check the consistency of the results, these are : I regularly check customer feedback to understand how out service are

performing (strongly agree to disagree), and how often do you read customer reviews about your company's products or services (Always to Never)

    - Accuracy: The survey results give correct/accurate insights into Talabat's specific requirements, expectations, and challenges expected because the ones who answered the survey have direct knowledge in the company and experience.

- Secondary Research : From multiple resources, articles, and reports, information was collected regarding AI technologies and sentiment analysis in e-commerce and their implementation, risks may happen, problems faced, accuracy, challenges in languages.
    - Accuracy: Using reviewed papers, and reports from IEEE verified improves the validity of the findings by these reports
    - Reliability: Many resources (papers, reports) are consistent, has the same main ideas about the benefits and the challenges of AI, confirming the findings reliability. Some technical topics were complex which led to misunderstanding If not carefully understood.
- Why the Selected Methodology is Best for Talabat and This Project
    - The survey addresses specific, practical data from its internal employees (managers, customer support, IT teams) what explains current expectations and their issues. Data from the survey, when integrated with the existing knowledge that exists from trusted papers about AI and sentiment analysis, ensures that the project can be done with providing real company needs and proven research. Purposive sampling ensures that the survey has been answered by people that in knowledge, and reaches the most relevant stakeholders, which is important given the company size and the focus is on customer feedback systems.
- Relation Between Methodology, Sample Size, and Sample Characteristics
    - The way the survey is done makes it limit the quantity of people answering the survey by targeting only specific roles within Talabat,

ensuring quality responses are meaningful to the project scope. This makes the findings important but may not reflect all the reflections on the system in the company. The secondary research provides the understanding of AI with sentiment analysis used in customer feedback systems globally. With these 2 methods that complete each other, the survey gives local specific data, while the secondary research gives a wider understanding of implementing the system.

- Reflection on Findings and Their Impact on Decision Quality
    - The survey results gave clear notes about budget, timeline, and functional needs, giving the roadmap for the project realistically. The survey also highlighted the concerns of data privacy, accuracy, and change management, which assisted in planning for mitigation strategies and risk planning. The secondary data highlighted the importance of implementing the AI sentiment analysis system and noted its common problems like the complexity of the Arabic language and giving value to the project and its approach for Talabat. These findings helped improve the accuracy of any decision made regarding the project by combining the real live data by the users with the external knowledge used by reports. Without forgetting that results should be integrated with caution, with continuous feedback from Talabat's team during the development which will help improve the accuracy.

2. Project Evaluation

Evaluate the planning recommendations made about the needs of the identified organization and the accuracy and reliability of the research carried out. Your evaluation may include how the recommended budget, timeline, risks, resources, and change management plan meet (a) the needs of the organization and (b) the accuracy and reliability of the research carried out.

A) How the Project Recommendations Meet the Needs of Talabat

- Budget: The recommended budget reaching up to JOD 50,000 is compatible with Talabat's expectations gathered from the managers in the survey. The budget includes software development, integration, testing, and staff training without the need for new hardware. This makes the plan financially logical and can be implemented for Talabat

Improvement: The budget is nearly as equal to the estimated costs, so it would be better to suggest a slightly higher amount in case unexpected actions happen or scope changes which often happen in AI projects.

- Timeline: The project timeline expected by managers for 4-6 months is appropriate for Talabat's needs, which gives time to test and deploy the AI system efficiently. By including clear milestones throughout the timeline (model ready, dashboards ready, testing complete, deployment, and project completion), makes sure that the project has good progress tracking and specified timely delivery.
- Risks: The project's risk management plan highlights key risks including incorrect sentiment analysis due to complexity of language, integration problems that exist within Talabat, data privacy issues regarding customer data, resistance from staff for new changes, and budget overruns. Each risk mentioned has its mitigation strategies, which show the preparation to handle any challenge that could occur and reduce their effectiveness on Talabat's operations.
- Resources: The plan for resources has many skilled employees including (AI developers, data engineers, frontend developers, testers, trainers) to each project phase, which matches Talabat's existing infrastructure and staff capabilities. With these resources, it will allow smooth implementation without any problems with Talabat teams that already exist.

B) Evaluation of the Accuracy and Reliability of the Research

- The primary research which is the survey used purposive sampling method which its main objective is to target Talabat's managers and employees that are involved with customer service and IT, which makes the survey containing relevant data that the company needs, budget expectations, and concerns.

- The survey included questions that made sure of reliability and consistency, with also two related questions about feedback monitoring by employees, and a question of the years of experience of the employee answering the survey, which improves the reliability of the data collected.
- Even though the sample size made the survey limited to number of people, the targeted sampling ensures quality data collected and importance.
- The secondary research included a review of verified articles and reports on AI and its integration with sentiment analysis and how its used in customer feedback, confirming the validity of the AI approach for Talabat's needs and highlighting main problems and solutions.
- By combining these 2 methods, it provided a balanced, accurate, and reliable knowledge base that supports the starting of project planning and decision making.

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Appendices